

Multiple Choice (1 point each)

Question 1

Pseudo code is a high level language for computers.

- ☐ True
 - ☐ False
-

Question 2

The worst-case time complexity of insertion sort is:

$$O(n \log n)$$

- ☐ True
 - ☐ False
-

Question 3

In divide and conquer algorithms, we break down a bigger problem into smaller ones, solve those smaller problems, and then combine the solutions.

- ☐ True
 - ☐ False
-

Question 4

The height of a complete binary tree of size n is approximately:

$$\log n$$

- ☐ True
 - ☐ False
-

Question 5 (2 points)

In a memory-limited system, which one of the following sorting algorithms would you use and why?

1. Selection Sort
2. Merge Sort

Answer:

Question 6 (2 points)

Which of the following sorting algorithms is most efficient for sorting arrays that are already mostly in order?

- ☐ Merge Sort
 - ☐ Selection Sort
 - ☐ Insertion Sort
-

Question 7

The formula for the sum of consecutive numbers starting from 1 to n is:

$$\frac{n(n-1)}{2}$$

- ☐ True
 - ☐ False
-

Question 8

The symbol for tight bound is:

$$\Theta$$

- ☐ True
- ☐ False